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Auto-Diagnostic Solution for Multiphase Flow Meter Systems. Case Study & Lesson Learned from an Oil Operator in Abu Dhabi

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Abstract

This paper offers a comprehensive exploration into the field applications of Multiphase Flow Meters (MPFM) across global contexts, and the insights from the lesson learned of a smart oil field that truly represents an exemplary sample for evaluating MPFM technologies, as it encompasses several types of MPFM available in the market. By delving into the practical implementations of MPFM technology, the study uncovers invaluable lessons learned and best practices derived from diverse operational environments that drove the development of a robust auto-diagnostic solution to enhance MPFM performance.

Through meticulous analysis and flow test data examination, this research illuminates the pivotal role of MPFM in optimizing oil field operations, enhancing production efficiency, and mitigating operational challenges imposed by subsurface complexity, CO₂ injection, gas lift operations, testing equipment reliability, configuration, and limitations. Additionally, the paper provides an in-depth examination of common pitfalls encountered in MPFM applications, offering guidance for successful implementation strategies into future projects. Furthermore, the author reviews several technical papers detailing MPFM implementations worldwide, integrating the outcomes of published flow loop tests to enrich the discussion. The know-how and best practices obtained in this research help to develop an agnostic & Automatic diagnostic solution for MPFM system, that oversees the real time data while testing the wells, providing a health check report, indicating the potential root cause of low-quality test from the testing equipment perspective, distinguishing between genuine changes in well performance, metering issues, limitation, etc.

Additionally, a benchmark study was conducted across numerous Oil fields, where some assets demonstrated acceptable MPFM performance while others did not. One Asset in particular faced significant challenges related to the quality of well tests post-MPFM installation, resulting in a high well test rejection rate, leading to a negative impact over the system efficiency, inefficient production scenario and poor reservoir management. However, these challenges were overcome after an exhaustive investigation, identifying the major root causes of errors that lead to the successful development of an automatic diagnostic system that support the process of well test quality assurance, highlighting equipment underperformance,

sizing evaluation, process upsets, radioactive source issues or decay, proactive detection of fluid properties deviation from "the references" or well performance related issues through a real-time framework.

The findings serve to inform industry stakeholders and researchers alike, offering valuable lessons learned from real applications about the adoption of MPFM technology, providing guidelines to achieve superior performance, data quality and reliability for operational excellence.

Introduction

The implementation of multiphase flow meters (MPFMs) represents a significant advancement in production monitoring and reservoir management for the oil and gas industry. By enabling direct, real-time measurement of multiphase fluids at the wellhead eliminating the need for conventional phase separation equipment, thereby reducing operational complexity, footprint, and associated costs. Their integration into both onshore and remote or subsea environments underscores their versatility and the strategic value they bring to operators seeking reliable and continuous well surveillance. Effective deployment of these meters, however, demands careful consideration of operational parameters, fluid properties, and measurement principles, as these factors directly influence data quality and system reliability.

Recent field studies, particularly those conducted across the fields studied, have demonstrated that the performance of MPFMs can vary widely depending on asset-specific characteristics and operational practices. Lessons learned from these applications highlight the necessity for robust auto-diagnostic capabilities and continuous calibration to address measurement uncertainties and maintain high standards of data integrity. The transition from traditional well-testing methodologies to MPFM-based frameworks has enabled more agile and cost-effective well-performance evaluations yet has also introduced new challenges in terms of equipment diagnostics, root cause analysis, and proactive identification of anomalies.

As MPFM technology continues to evolve, the development of advanced health monitoring and diagnostic solutions has become imperative. These tools provide operators with actionable intelligence, facilitating prompt response to correct deviations, equipment malfunctions, or process upsets. Furthermore, benchmarking efforts across multiple assets reinforce the importance of systematic implementation strategies, including well test quality assurance procedures, equipment sizing evaluations, and real-time performance tracking. Through these measures, operators can maximize the value of MPFM investments, achieve superior operational efficiency, and ensure the long-term sustainability of smart field initiatives.

A benchmark study across several onshore assets revealed one asset in particular experiencing high well test rejection rates after MPFM installation, resulting in inefficiency and poor reservoir management. After an in-depth investigation, the sources of errors were identified (Ref1.); and an auto-diagnostic tool was developed to automatically detect MPFM issues integrating real time equipment's alarms, ensuring well test quality and identifying equipment or fluid property problems on spot. This solution increased test acceptance rates from 50% to 85%, reducing discrepancies with portable separators, and significantly improved system efficiency and production performance.

Multiphase Flow Meter Global Applications

Multiphase flowmeters are being adopted by Oil and gas Operators to measure the production fluids without any need of phase separation. Many operators have incorporated them in their testing facility design based on their smaller footprint and high reliability, especially in remote areas or subsea applications. They are also attracted by the ability to measure all three phases in continuous bases, when they are connected within a real time framework along with the competitive cost versus fix separators that usually require additional pipping and equipment leading to high capital expenditure (CAPEX) along with operational upsetting on downstream facilities. However, when MPFMs are chosen over Three-phases test separators there is a tradeoff, compromising rate estimation accuracy.

Multiphase flow meters also present challenges. They need to cope with a wide range of variable parameters and flow conditions, such as flow pattern, gas void fraction (GVF), water cut, pressure, oil density, oil viscosity, and water salinity changes among others. Different meter conceptions exhibit varying characteristics depending on the physical principles selected for measurements, leading to different envelopes of application and measurement uncertainties. Despite achieving adequate performance for selecting applications, careful flow condition analyses remain essential.

Table 1—Types of Multiphase flow meters.

Type	Measurement Principle	Strengths	Limitations	Best Use Cases
Differential Pressure (DP)	Venturi + pressure drop correlations	Simple, robust, good for high-pressure environments	Less accurate at high GVF ; needs phase fraction inputs	Onshore, standard multiphase wells
DP + Gamma-Ray (nuclear)	Gamma attenuation for phase density/fraction	High accuracy in complex flows, good for high GVF	Radioactive source; HSE regulated, Application limited by venturi size	Subsea, high GVF, deepwater, mature wells
DP + Capacitance/ Dielectric	Measures dielectric constant of phases	Fast response ; good water/oil cut measurement	Affected by emulsions salinity. Application limited by venturi size	Water-cut monitoring
DP + Conductivity Based	Measures electrical conductivity	Accurate water detection in saline systems	Ineffective with low salinity or emulsions. Application limited by venturi size	High-water-cut wells, especially with known salinity
Ultrasonic	Doppler/transit time +speed of sound	Non-intrusive ; no moving parts; real- time velocity	Accuracy drops with slugging or gas bubbles	Clean production
Coriolis	Tube vibration to measure mass flow and density	Direct mass measurement; high accuracy in steady flow	Not suitable for high GVF or slug flow	Well testing with low GVF
Optical/Infrared	Light absorption / reflection per phase	High-resolution, phase-specific	Sensitive to contamination	Laboratory setups; clean fluids
Magnetic resonance	Fluid magnetic field	Non-intrusive, no moving parts, no limitations with pipping. /direct fluid characterization.	Few field applications (lab – scale)	Good in low salinity environment

Flow Loop Tests

Recent research demonstrates that MPFM are capable of operating at high gas volume fractions (GVF) in the range of 85-99%; however, accuracy degrades significantly under these conditions. A review over thirty technical papers covering real field applications across diverse operating regions (including KSA, UAE, Alaska, USA, Ecuador, Russia, Gabon, Canada and Brazil) confirms that measurements errors range from 15-40% for the main phase (Oil) at high GVF. In this paper, the author selected two pieces of work that were the most relevant to the studied fields and referred to those studies as examples.

Figure 1, adapted from Reference 2 (Costa et al., 2008), illustrates the observed discrepancies in liquid measurement between three-phase separators and Multiphase Flow Meters (MPFMs) utilizing dual gamma technology under high Gas Volume Fraction (GVF) conditions, particularly in gas-lifted wells. Costa et al. attributed these liquid rate deviations to the turbulent flow fluctuations that occur within the venturi section of MPFMs; phenomena that may not be adequately captured by traditional three-phase separators.

Despite these findings, operational experience in the fields studied suggests that the elevated liquid rates recorded by MPFMs are not always consistent with the established production profiles of individual wells. In many instances, these discrepancies are inconsistent with the volumes recorded by custody transfer meters at field level, ultimately diminishing overall system efficiency and increasing production allocation errors. Additionally, irregular fluctuations in water cut (WC) and gas-oil ratio (GOR) were observed that do not correspond to expected physical behavior. Moreover, Real-time wellhead data showed that differential pressure fluctuations were noisier and wider than those seen by wellhead flowing temperature and pressure transmitters and did not align with slugging patterns. In contrast, both instant and cumulative rates from the 3-phase separator closely matched the well flow at the nearby wellhead as shown in Figure 2.

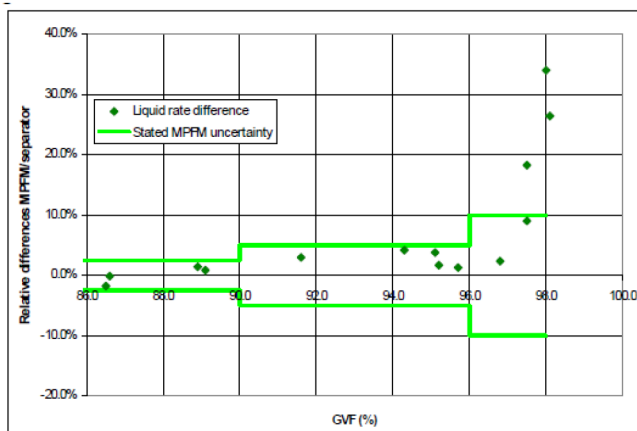


Figure 1—Shows liquid differences between separators and MPFM increased at high GVF.

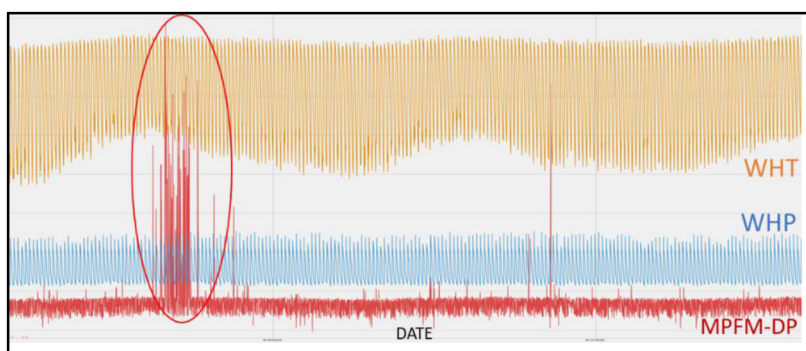


Figure 2—MPFM differential pressure is compared with wellhead real-time data, showing an increase in differential pressure while flow conditions remain unchanged.

Ali Faraj et al. (2018) observed that while multiphase flow meter (MPFM) technologies have advanced, certain challenges persist; particularly with respect to accurately measuring continuous well rates under challenging flow regimes, such as slug flow or conditions with elevated gas-oil ratio (GOR) and gas volume fraction (GVF). Their flow loop experiments evaluated three MPFM models, however, some limitations were nonetheless identified (Figure 3).

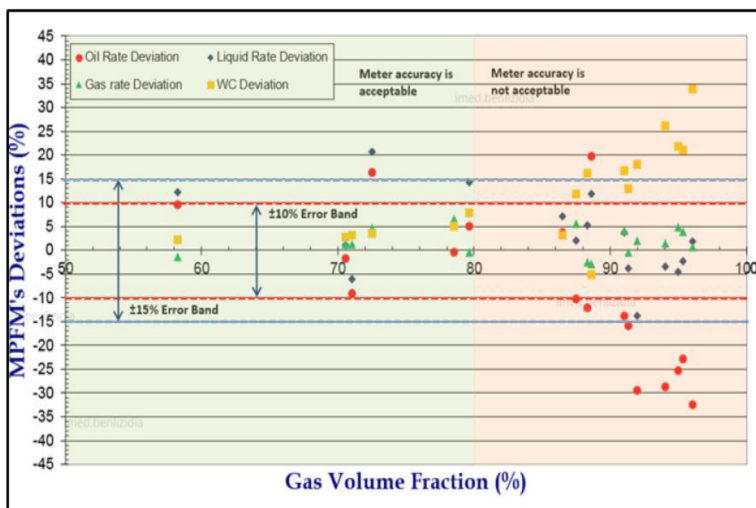


Figure 3—Real Example of a flow loop test designed to evaluate MPFM performance in High GOR wells (WC & Oil off)

Field Background & Operational Environment

The Studied fields offer diverse types of production environments that can prove challenging for effective production well testing with conventional Multiphase flow meters. The fields fall into three different categories:

1. Some reservoirs are light oil, low productivity index with high water cuts and low gas-oil ratio. Moreover, the wells are assisted with gas-lift, causing a challenging operational environment for most of the multiphase flow meters in the market to work at high gas volume fractions (GVF) >98%.
2. Other group of wells flow naturally, exhibiting high GOR between 5000 to 12000 scf/Bls having similar GVF range as per the first group with the aggravating of high content of H₂S or CO₂.
3. Finally, a group of wells with medium to low GOR using ESP or gas lift as artificial lift methods with scale deposition tendency and compositional gradient that imply the use of a complex fluid model to cope with the significant fluid properties variation vertically and areal across the reservoir.

In addition all reservoirs are under EOR programs either water, CO₂ or Water alternate Gas injection (WAG) in small patterns, therefore water or gas breakthrough is imminent, impacting the accuracy of the MPFM, depending on the injectors water or gas mode scheme changing salinity, density and gas void fraction (α). This wide range of operational condition create challenges when designing MPFM's & analyzing their performance, affecting well and network models prediction and asset decision making for overall field production enhancement.

Before the commissioning of the Multiphase Flow Meters (MPFM) and Gas Lift (GL) meters in 2018, the assets faced challenges. The newly commissioned wells had no production history, leading to multiple sources of errors. Consequently, PTS test verification was required to ensure accuracy. However, after the implementation of MPFM and GL meters, there was a significant drop in well test acceptance. The MPFMs were highly dependent on PVT parameters, making them less suitable for miscible EOR projects in reservoir with compositional gradient. This dependency demanded frequent Portable test units (PTS) and sampling campaigns to validate changes in fluid properties. The actual differences found during PTS verification ranged from 2% to 100%, with 70% of strings operating under gas lift, causing MPFMs to work at the boundaries.

Impact of Well Test Errors on System Efficiency and Reservoir Management

Well test inaccuracies in the fields studied have been identified as a critical factor adversely affecting system efficiency. High well test rejections can result in substantial misallocations, leading to significant deviations in production and recovery estimations, impacting the most the small reservoirs; these inaccuracies can translate into Recovery Factor errors of up to 30%, primarily due to erroneous back-allocation. In addition, the low frequency of well testing exacerbates these issues by introducing anomalous and inconsistent production profiles, which can distort reservoir performance assessments. When well test data, subject to substantial uncertainty, is employed for production allocation, the resulting errors propagate through reservoir evaluation models, adversely impacting reserves estimation, production forecasting, and decision-making processes. These discrepancies not only reduce production efficiency but also undermine the reliability of development plans. The lack of accurate, timely, and representative well performance data limits the operator's ability to implement effective reservoir management strategies, thereby increasing operational risk and constraining field optimization potential.

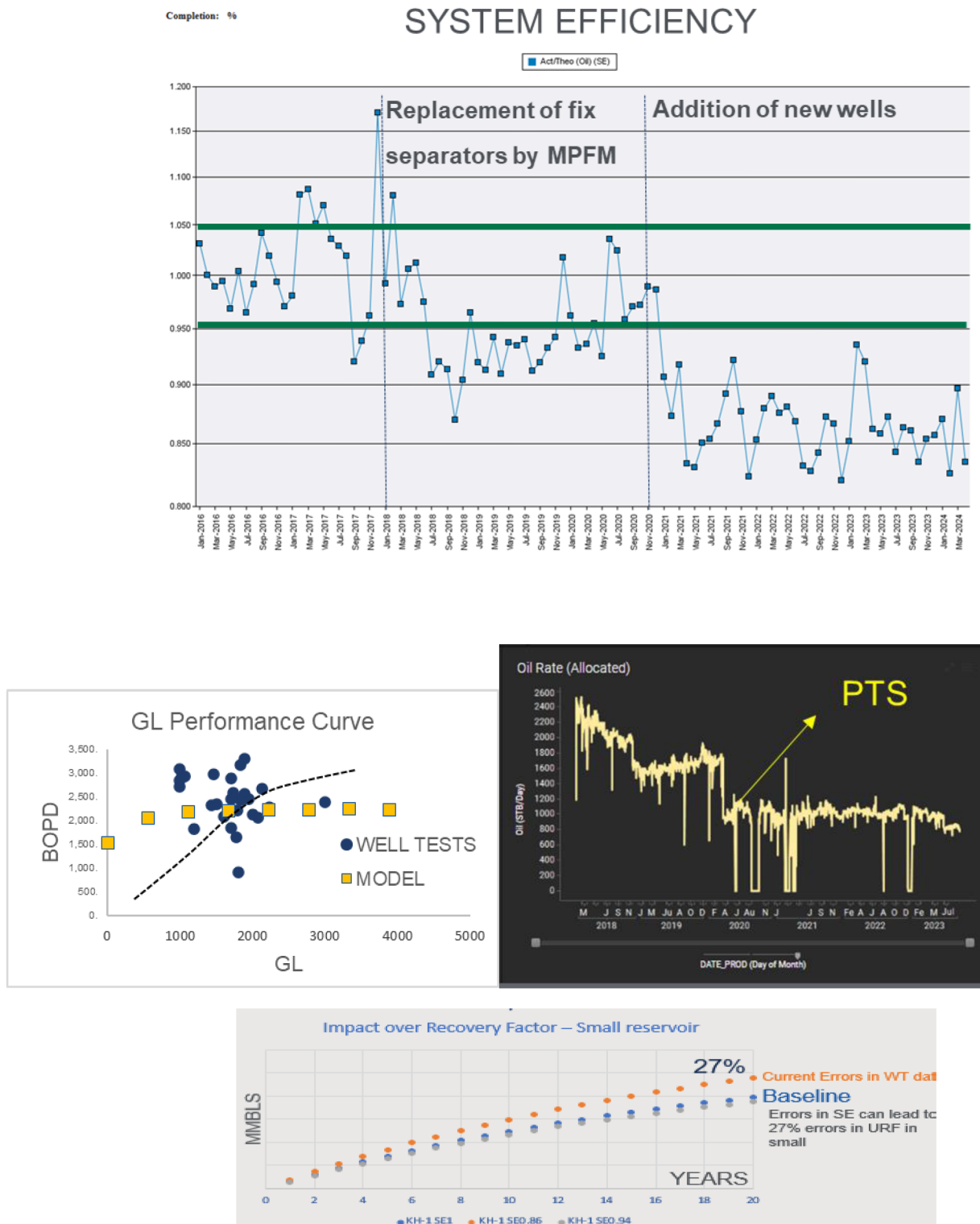


Figure 4—Examples of impact of Well test errors and low frequency in reservoir management

Challenges & Observations after implementation of MPFM

1. Dependency on PVT Parameters: MPFMs are highly dependent on PVT parameters, making them less suitable for miscible EOR projects and requiring frequent PTS and sampling campaigns to validate changes in fluid properties.

2. Additional cost: requirement of additional Portable test separator Units (PTS) to validate MPFM performance to correct allocation.
3. Vendor-Dependent Interventions: MPFM interventions were highly vendor-dependent, making it difficult to maintain high reliability without vendor support.
4. Oil Rate Overprediction: Issues related to oil rate overprediction were observed, with discrepancies in oil rate predictions highlighted in the data.
5. Abnormal GOR and High GOR Variation: Well test issues related to total gas volume less than gas lift or abnormal GOR with high GOR variation overtime.
6. Process-Related Issues: Issues such as passing valves or wells tripping while diverting to testing facilities.
7. Low Oil and High GVF: Challenges in measuring formation gas and discrepancies in oil rate measurements were observed.
8. MPFM Malfunctions: half of the MPFMs had no improvement after recalibration, indicating a low success rate in MPFM intervention.
9. MPFM Design limitations: The majority of MPFM venturi sizes are determined based on well production profile forecasts prior to the commencement of production. No single size can accommodate the full range of well productivity throughout the life of a well or within the same cluster.

Lesson Learned & Best Practices

MPFMs offer operational efficiencies and cost advantages in offshore and subsea production by enabling continuous flow rate measurement, particularly when elevated H₂S levels pose risks from a production handling standpoint. The technology can now deliver measurement uncertainties below 5–15% under most conditions, except for high GVF and slug flow. Moreover, there are specific scenarios, such as high water cut with low GOR and high gas lift injection that require more precise (<5%) and redundant meters with real-time monitoring to address uncertainty.

During the MPFM design stage, it is important to consider the complete production profile of the wells and potential production decline, which may require adjustments in venturi size over time. When using MPFM in onshore applications, for multiple wells in a cluster or at a flow station, it is advisable to avoid DP-based MPFM to prevent wells from operating outside the equipment's optimal range. If avoiding a DP meter is not feasible, it is recommended to install dual venturi MPFM or use more than one MPFM to accommodate changeable productivity levels and well performance. Another option is to request MPFM that can easily detach the venturi module and replace it with a suitable size. Ensure that the sample points provision is aligned closely with the MPFM, particularly at the gas stream in cases involving miscible gas or CO₂ injection.

It is recommended to perform MPFMs verification with portable units at least every year, especially in reservoir under EOR programs with complex fluid models expecting fluid properties variation over time. In onshore fields, Truck mounted portable MPFM was key to accelerating equipment mobilization across clusters having a positive impact, doubling well test frequency.

Find below a list of mitigations considered in this case study that helped to improve well test quality:

Low MPFM Reliability

Challenge. The reliability of Multi-Phase Flow Meters (MPFMs) was low, leading to a high dependency on Portable Test Separators (PTS) units. This results in additional costs and operational inefficiencies.

Mitigation.

- Select the right MPFM to cope with specific operating conditions.

- Introduce flow loop tests in the MPFM screening process to measure the expected accuracy and its impact over reservoir management.
- Emphasize the utilization of technologies designed to operate effectively under the conditions of High Gas Volume Fraction (HG VF) and high water cut. In such scenarios, it is recommended to implement systems equipped with gas conditioning (partial separation) capabilities as well as redundant water cut meters.

High Vendor Dependency

Challenge. Dependency on MPFM vendors for field interventions and troubleshooting may result in delays to MPFM activities, potentially affecting both the frequency and quality of flow testing.

Mitigation.

- Upskill the internal taskforce to configure new well profiles in MPFMs. Collaborate with MPFM providers to integrate the equipment in the real time network to be able to make modifications in the well profiles remotely.
- Promote the creation of a centralized department to maintain MPFM reliability and troubleshooting.
- Increase PTS units to support yearly MPFM verification. Validate the fluid model with single-phase samples, which helps reduce external lab costs for PVT analysis and speeds up resolving discrepancies using portable units.

Gas Lift Sensitivity

Challenge. Gas lift sensitivity over fluid density is not captured in MPFM fluid profiles, leading to errors in oil rate predictions.

Mitigation.

- Update the PVT table for each well at the optimal gas lift rate as needed.

Manual Data Entry

Challenge. Manual data entry is prone to human errors and slows down the well test data records in official databases for well test validation, impacting testing accuracy & frequency.

Mitigation.

- Realtime data and automation of the well test validation process.

Lack of Proper Sample Points

Challenge. There is a lack of proper sample points to update fluid properties in MPFMs, leading to inaccuracies.

Mitigation.

- Incorporate WC/H₂S/CO₂ analyzers for redundancy to confirm MPFM results and reduce emissions and workforce associated with manual sampling.

Budget Constraints

Challenge. Budget constraints limit the ability to perform necessary MPFM interventions and upgrades.

Mitigation.

- Develop an auto-Diagnostic MPFM Alarm system to optimize costs and adjust the MPFM maintenance contract strategy, duration, and funds based on business requirements.
- Develop a tracking sheet per equipment to build a meaningful database to build a model to predict failures.

Facility Design Issues

Challenge. Facility design issues, such as MOV/LCV problems and wells tripping on high pressure, impact testing processes.

Mitigation.

- Replace problematic MOV/LCV components.
- Revisit well trip points and resolve issues for most wells.

Auto Diagnostic Solution

Well test measurement errors can result from a complex combination of factors spanning equipment performance, flow dynamics, and well behavior. The most significant contributor to well test rejection (>50%) is related to MPFM/FTS equipment malfunction or limitations, such as poor calibration, fluid model mismatch, inadequate meter sizing or range, gamma sensor instability, communication failures, or piping constraints (Figure 5). These equipment-related issues often go undetected without a robust diagnostic system and can severely compromise data quality. Additional contributors include unstable flowing conditions (>25%), influenced by factors like flashing, stabilization time, backpressure, or valve malfunctions. Well performance issues such as gas breakthrough, GOR/WC rate dependency, and gas lift inefficiency also account for more than 20% of deviations. The path-to-surface correction—typically involving volume and phase adjustments has the least impact (<5%). These multivariate sources of error underscore the need for a comprehensive anomaly detection system to improve reliability in well testing since unavailability of MPFM diagnostic data leads to late detection of faults during well testing. These alarms include:

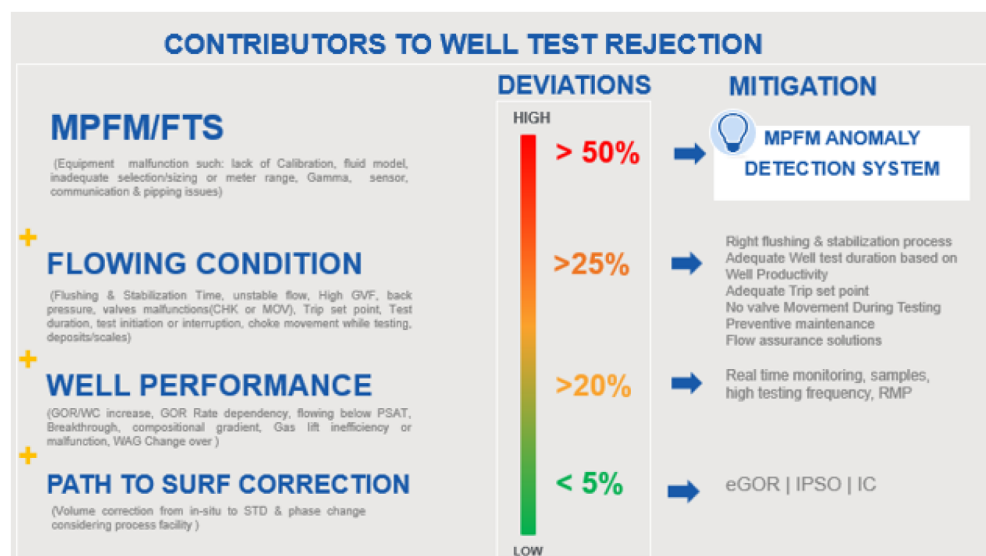


Figure 5—Contributors to Well Test Rejection

The **New MPFM Auto-Diagnostic system** was tailored as per device manufacturer and is captured through advanced DCS configuration and special interface for reporting to improve awareness of meter performance. It executes real-time meter health checks, addressing over 50% of well test rejection errors in the well test process. The implementation of the Anomaly System has significantly enhanced the MPFM performance and increased the success in the calibration process. Before the system was in place, identifying and addressing issues with the MPFM was a reactive process, often leading to delays and inefficiencies. However, with the new system, we can now proactively identify problematic MPFMs, accelerating troubleshooting and optimizing resource allocation. This proactive approach ensures that we maintain high performance and reliability in the well testing operations. Also, it helps to eliminate the bias against MPFM and distinguish between equipment issues and real changes in well performance.

Figure 6 shows the MPFM Test report that provides detailed production data collected during a well test operation, both at standard conditions (SC) and actual conditions (AC). The data set captures essential flow parameters and supports quality control through the integrated alarm detection system in the bottom right corner. The following Key Operational Parameters are input to the Auto-Diagnostic system:

- Instant/ Cumulative Real time Oil Flow (SC/AC)
- Instant/ Cumulative Real time Gas Flow (SC/AC)
- Instant/ Cumulative Real time Water Flow (SC/AC)
- Average GOR
- Average GVF
- Flow Pressure
- Flow Temperature
- Mixture Density
- Conductivity of Flow: 0.00 (suggesting no salinity or sensor misread)
- Water Cut (SC): 0.41% (indicating extremely low water production)

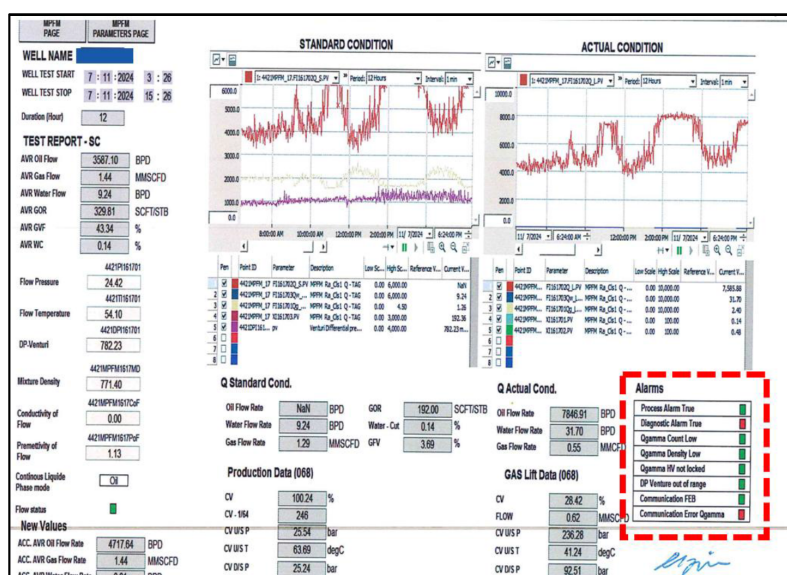


Figure 6—Well test report sample highlighting Auto diagnostic system. This example shows two alarms open: diagnostic and gamma communication error.

Alarm System – Quality Assurance Layer

The MPFM system features a comprehensive alarm diagnostics panel, which provides real-time quality assurance through the Auto-Diagnostic solution. This report details the alarms and indicators as follows:

Process Alarm

The MPFM system is equipped with a sophisticated suite of process alarms designed to safeguard measurement accuracy and operational safety. These alarms are configured to trigger in response to both high and low excursions of venturi differential pressure, functioning as an early-warning mechanism for potential issues such as improper venturi sizing or dynamic changes in flow conditions.

The system also continuously monitors a range of critical process parameters, including water cut, mixture density, and the physical properties of both oil and water—such as conductivity, permittivity, and salinity. Process alarms are activated whenever these indicators display abrupt deviations or sustained fluctuations, which may signal instrument drift, calibration errors, or shifts in fluid composition.

Particular attention is given to Gas Volume Fraction (GVF) levels, with alarms set to alert operators if readings exceed 95% or fall below 25%. These thresholds are crucial for maintaining the integrity of multiphase measurements, as extreme GVF values can destabilize sensor performance and compromise the reliability of flow data.

Furthermore, the system tracks flow-pressure and temperature metrics, flagging any instance where values deviate beyond the established Pressure-Volume-Temperature (PVT) parameter ranges. If PVT data exceeded known physical limits or reference values; for example, measured density falls below the expected density of gas; then, an alarm is generated. Such comprehensive surveillance ensures that any anomaly, whether due to sensor malfunction, process upset, or abnormal fluid properties, is promptly detected and addressed.

These process alarms, functioning in concert with diagnostic and communication alarms, create a multilayered defense that underpins the reliability and quality assurance of MPFM well test results, thereby supporting robust decision-making and operational efficiency across diverse field deployments.

Diagnostic Alarms

This alarm is designed to monitor the operational health of the equipment. It is activated in scenarios such as empty pipe offsets, where flow detection falls outside expected parameters, as well as in instances of sensor readings exceeding permissible operational ranges. In addition, the diagnostic suite includes watchdog status warnings to signal irregularities in system processing, notifications of potential hardware malfunctions, and clipping errors that may occur due to frequent or rapid shifts in the electrical sensor's operating modes.

Gamma Count Low, Gamma Density Low, Gamma Not Locked

In addition to the previously mentioned alarms, the system also identified several gamma module-specific health concerns, including Gamma Count Low, Gamma Density Low, and Gamma Not Locked. These alarms are indicative of issues such as the gamma detector operating outside of its designated temperature parameters or the module's inability to achieve or maintain a locked, stable measurement state. Such anomalies can significantly affect the accuracy and reliability of multiphase flow rate measurements, highlighting the necessity for ongoing monitoring of detector performance and temperature stability within the MPFM system. By promptly identifying these conditions, the system ensures that any deviation associated with gamma measurement is addressed efficiently, thereby maintaining the integrity and quality of well test data. These issues require vendor intervention, and the system facilitates timely resolution while optimizing vendor visits and associated costs.

DP Venturi Out of Range

Continuously monitoring of the differential pressure across the venturi is fundamental to assess the quality of the test, with dedicated alarms configured to detect both high and low cases. These "DP Venturi Out of Range" alarms along with real time trend of the differential pressure during the test, serve as early indicators of conditions such as improper venturi sizing or unexpected shifts in process dynamics that may compromise measurement accuracy. These alarms allow personnel to promptly set appropriate mitigation measures. For instance, if a well exhibits low differential pressure (DP), an additional test may be scheduled with an increased gas lift rate to raise differential pressure within the optimum DP range, improving the quality of the test. Conversely, if the DP is excessively high, adjustments such as reducing the production choke or gas lift rate may be considered on a case-by-case basis to achieve more accurate test results. As a result, test frequency and acceptance increased.

Communication FEB/Gamma

Communication anomalies and report generation issues are also monitored, specifically those related to the Front-End Board (FEB) and the Gamma module. Communication alarms are triggered when there are interruptions, failures, or inconsistencies in the data exchange between critical hardware components such as: transducers, sensor modules, or the gamma detection system. These alerts play an essential role in the early identification of issues that could disrupt the integrity of real-time measurements or compromise the synchronization of critical system parameters. By continuously overseeing the communication pathways within the MPFM architecture, the system ensures that any loss of connectivity, data corruption, or module misalignment is promptly detected and addressed. This layer of monitoring further enhances operational reliability, supporting proactive maintenance and minimizing the risk of undetected data transmission faults that could affect well test accuracy.

Figure 7 shows how the automated anomaly detection system helped identify and resolve equipment issues that had gone unnoticed during four years of MPFM interventions. Following these fixes, the data is now more consistent with PTS results and no alarms are active, confirming the integrity of the intervention outcomes.

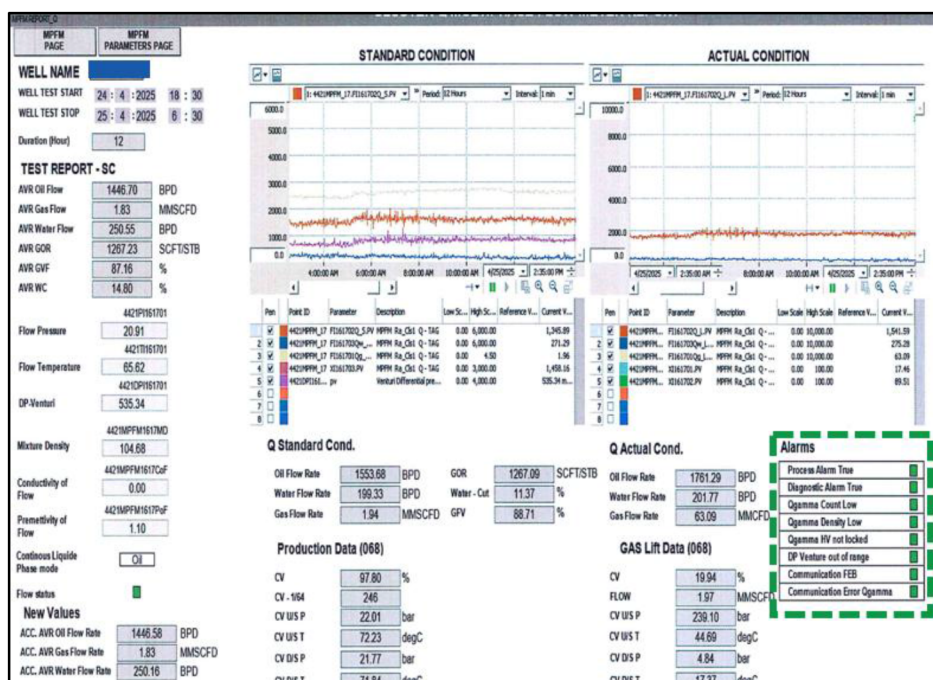


Figure 7—Well test report including MPFM Auto-Diagnostic Alarm system after successful intervention to clear active alarms.

Results

The MPFM auto-Diagnostic system played a key role in identifying the root cause of flow discrepancies, allowing timely intervention and validation of results. With all critical flow parameters streaming in real time and integrated with the pre-set alarms cleared in the final well test report, provides a robust, field-ready framework for ensuring well test data quality across all MPFM deployments.

The realized value of the MPFM Auto-Diagnostic system is as follows:

- Significant improvement in MPFM performance and calibration's success.
- Facilitate Well test Quality check, distinguishing from well and equipment related issues.
- Proactive MPFM monitoring from the equipment standpoint, Accelerate troubleshooting.
- Reduce OPEX by having more effective intervention planning.

Figure 8 It is an independent source of verification of the actual improvement in well test quality at field level. The system efficiency factor that is the ratio between actual rate from export meters and the totalization of the allocated rates based on well tests had increased from 0.86 to a remarkable 1.02 due to the successful mitigation plan implemented by the asset to improve meter reliability and accuracy.

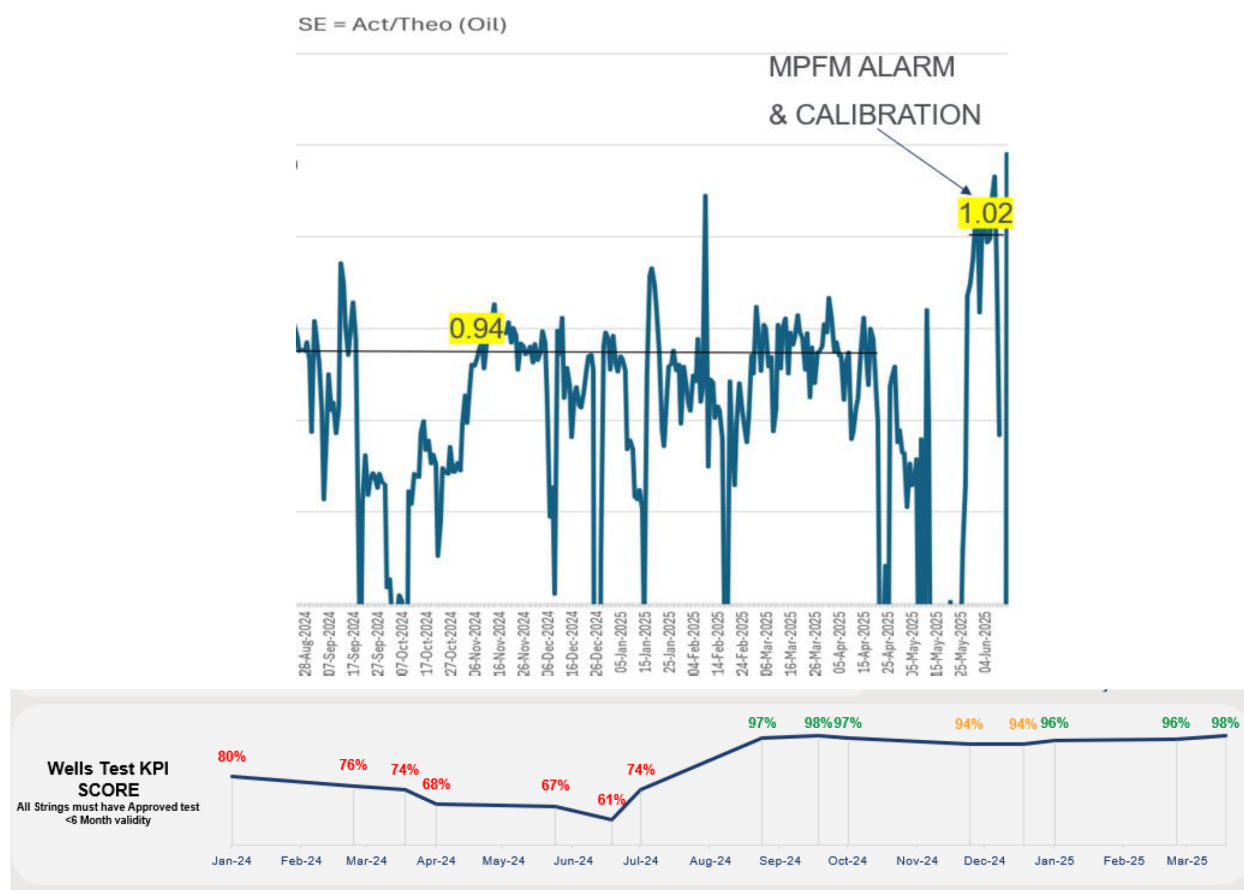


Figure 8—Improvement in field efficiency factor due to the implementation of the MPFM Auto-Diagnostic system.

Conclusions

- This highlights the necessity for comprehensive calibration and validation protocols, especially when deploying MPFMs in environments characterized by high GVF and dynamic operating

conditions. Continuous monitoring, advanced auto-diagnostics, and rigorous cross-verification with portable separation units or custody meters are essential to ensure reliable measurements and to maintain the integrity of production data across the asset portfolio.

- The MPFM auto-diagnostic solution significantly improved flow-test quality, reducing OPEX and inefficiencies in reservoir management. The optimization of artificial lift and reservoir management is strongly affected by flow rate errors and the low frequency of well tests.
- It is important to set clear expectations on the accuracy of the measurements that are expected based on the testing equipment and the following condition that would be acceptable to drive reservoir management. For exceptional cases like miscible injection floods, it is important to understand the potential impact on the physics of the measurement device as well as any reference device that may be involved.
- All multiphase devices require periodic calibration to cope with fluid evolution and changes its fluid properties. Therefore, the deployment plan should include provisions to acquire representative produced fluid samples as close as possible to the testing equipment to optimize additional cost of conducting PVT and lab analysis to properly set up a representative fluid model into the multiphase flow meter.
- When attempting to obtain accurate oil, gas and water rates in mature fields with high water-cut and gas rates in slugging wells, it is desirable to have single phase properties from sampling, along with high frequency real time data connected to SCADA to improve the gas lift rate estimation using totalizers to calculate the cumulative gas lifted volume during the test to minimize significant errors in GOR. Also, redundant metering and high frequency tests to overcome meters accuracy.
- It is critically important to train the operational personnel to know the principle of operations for the multiphase devices being used, so they can diagnose signs of malfunctions and report the problems encountered.
- Most multiphase measurement devices have their own formats for reporting the data output desired by the operators. These formats may not be compatible with the format requirements established by the operator's SCADA system. This issue must be examined, corrected and tested before the formal operation is initiated.
- Keeping historical record of the problems encountered, remedies, and normal maintenance are important to establishing the sustainability of the multiphase devices and cost effectiveness of the technology.

Way forward

1. **Implement Well test digitalization strategy:** enabling remote Testing & monitoring, extend implementation of the MPFM Auto-Diagnostic solution to all fields, developing further AI-based models to access well test quality from the well performance perspective along with the equipment health check.
2. **Incorporate online WC and/or gas analyzers for redundancy** to confirm MPFM results and proactive troubleshooting, as well as reducing emissions and workforce associated with the manual sampling approach.
3. **Promote the creation of a centralized department** to support Operations in MPFM reliability/troubleshooting and to maintain proper update of the fluid models.

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Nomenclature

BHFP	: Bottom hole Flowing Pressure
CAPEX	: Capital expenditure
CO ₂	: Carbon Dioxide
DP	: Differential pressure
EOR	: Enhanced oil recovery
ESP	: Electric submersible pump
GL	: Gas lift
GOR	: Gas Oil Ratio
GVF	: Gas volume fraction
H ₂ S	: Hydrogen Sulfide
HGVF	: High Gas volume fraction.
MPFM	: Multiphase flow meter
OPEX	: operating Expenditure
PTS	: portable test unit
PVT	: Peressure-Volume_Temperature Analysis
THP	: Tubing Head Pressure
WAG	: Water alternate Gas injection
WC	: Water cut
WT	: Well test
α	: Gas void Fraction

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